

EMMANUEL MOBOLAJI ARO

SENIOR SOFTWARE ENGINEER · SYSTEMS & PLATFORM ARCHITECTURE · APPLIED AI

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LinkedIn • GitHub • Portfolio

EXECUTIVE PROFILE

Results-driven Senior Full Stack Software Engineer with expertise in architecting high-throughput distributed systems, scalable backend platforms, and applied machine learning solutions. Master of the Python (Django/Django Ninja) and TypeScript (Next.js App Router) ecosystems, with a track record of leading cross-functional engineering teams to ship world-class, production-grade platforms. Strongly oriented around clean domain-driven design, defensible API contracts, strict typing, automated CI/CD pipelines, and cloud-native containerized infrastructures that maintain sub-second latency and high availability under demanding production workloads.

TECHNICAL EXPERTISE

Core Languages	Python, TypeScript, JavaScript (ES6+), SQL, PHP, Bash / Shell Scripting
Backend & Cloud Architecture	Django Ninja, Django, Node.js, RESTful APIs, GraphQL, Microservices, JWT/OAuth2 Auth, Event-Driven Systems
Frontend & UI Engineering	Next.js (App Router), React, Tailwind CSS, Server Components (RSC), State Management, Web Performance / SEO
Data Engineering & Storage	PostgreSQL, MySQL, SQLite, Supabase, Redis, Schema Modeling, Query Optimization, ORMs & Migrations
DevOps & Cloud Infra	Docker, AWS (EC2, S3, RDS, IAM), CI/CD Pipelines (GitHub Actions), Linux Administration, Nginx, Postman
Applied AI & Machine Learning	Scikit-learn, TensorFlow, Pandas, NumPy, OpenCV, NLP Pipelines, Predictive Modeling, LLM API Integrations
Engineering Leadership	System Architecture Design, API Contract Design, Security Hardening, Agile/Scrum Leadership, Code Review & Mentorship

PROFESSIONAL EXPERIENCE

Boxonia Blueprint

Oct 2024 – Present | Lagos, NG

Software Engineer (Platform & Ecosystem Architecture)

- Architected and developed a centralized digital ecosystem in Python that consolidates story-development and talent-management workflows across 5+ globally distributed film production teams.
- Engineered an automated talent-booking and cross-border coordination platform, replacing multi-day manual scheduling loops with a zero-friction transactional flow that accelerated client onboarding velocity by 40%.
- Designed scalable relational schemas and secure RESTful API integrations, ensuring high-availability data synchronization between distributed production units under strict SLA requirements.

Tech4mation (NativeTalk)

Feb 2026 – Jun 2026 | Lagos, NG

Senior Software Developer, Platform Engineering

- Spearheaded backend architecture and feature delivery across core enterprise products, collaborating closely with Product and Engineering leadership to establish record-breaking sprint delivery velocity within the company.
- Architected **TaskForge Pro**, a high-throughput enterprise team-management platform powered by Django Ninja, Next.js (App Router), and PostgreSQL, utilizing a decoupled microservice boundary and strict typed API contracts.
- Hardened platform security posture by implementing robust JWT authentication, role-based access control (RBAC), and automated Postman API regression test suites, achieving 99.9% endpoint reliability prior to staging promotion.
- Refactored core messaging pipelines and backend logic in the *omnichannel-crm* service, decoupling channel adapters from core processing loops to support horizontal scaling under surge traffic.
- Standardized engineering environments by introducing containerized Docker workflows, eliminating environment drift and reducing onboarding time for new engineers from 3 days to under 2 hours.

Bincom Dev Center

Sep 2025 – Jan 2026 | Lagos, NG

Full Stack Developer

- Architected high-performance frontend interfaces using Next.js (App Router) and React, leveraging Server Components and optimized data-fetching patterns to reduce First Contentful Paint (FCP) by 35% and boost SEO rankings.
- Overhauled relational MySQL database schemas and refactored hot-path SQL queries, accelerating complex data retrieval by 20% and eliminating latency spikes under peak user load.
- Mentored junior developers in TypeScript best practices, modular component design, and git branching standards, elevating overall team code quality and delivery velocity.

Cyberbuddies.com

Apr 2024 – Aug 2024 | Ibadan, NG

Full Stack Developer

- Delivered end-to-end full-stack web applications from initial system design to production deployment, introducing continuous integration and automated testing pipelines that increased delivery velocity by 25%.
- Spearheaded the migration of production workloads to AWS cloud infrastructure, optimizing compute and storage resource allocation to reduce cloud operating costs by 18% while upholding 99.9% uptime SLA targets.

SafeSpace.org Association

May 2023 – Feb 2024 | California, US (Remote)

Operations QA & Automation Engineer

- Designed and deployed Python and Django automated review pipelines that replaced manual verification loops, increasing organizational operational throughput by 30% with complete deterministic auditability.
- Instituted rigorous data quality and automated testing protocols in containerized Docker environments, reducing data-entry error rates by 50% and establishing company-wide benchmarks for system integrity.

Fiatci.io

Sep 2022 – Mar 2023 | Brussels, BE (Remote)

Lead Full Stack Engineer

- Led a cross-functional engineering team of 7 developers through the end-to-end architecture, development, and production deployment of the company's flagship web platform — delivering the first successful commercial product launch in company history.
- Owned technical direction across frontend, backend, and DevOps infrastructure; established rigorous code review standards, CI/CD release strategies, and zero-downtime production cutover protocols.

APPLIED AI & MACHINE LEARNING SYSTEMS

Enterprise Malware Detection & Classification Engine

PYTHON · SCIKIT-LEARN · PREDICTIVE SECURITY ML

Designed and deployed a Python-based security analytics platform applying supervised machine learning to real-time endpoint malware classification. Structured with automated feature extraction and retraining pipelines for evolving threat corpora.

Real-Time Computer Vision Attendance Platform

PHP · JAVASCRIPT · OPENCV · RELATIONAL STORE

Engineered a high-speed biometric attendance system integrating webcam image streaming with OpenCV computer-vision pipelines. Automated real-time facial capture, normalization, and identification against a secured relational store.

Clinical Oncology Risk Prediction Model

SCIKIT-LEARN · PANDAS · EXPLAINABLE AI (XAI)

Developed an explainable, high-accuracy diagnostic classification model trained on clinical patient datasets for proactive risk assessment. Emphasizes reproducible feature engineering and interpretable outputs for clinical decision support.

NLP Misinformation & Named Entity Extraction Engine

NLP · TOKENIZATION · INFORMATION RETRIEVAL

Shipped end-to-end NLP pipelines in Python leveraging statistical embeddings and tokenization primitives to categorize unstructured text, extract critical named entities (NER), and detect disinformation in high-noise social data streams.

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science — Federal University Oye-Ekiti, Nigeria

2025

Advanced Software Engineering Certification — GB-Tech Learning Centre, Nigeria

2023